Sarvesh Nand Kumar Khetan

San Francisco, CA | +1 240 726 8367 | skhetan@terpmail.umd.edu | Linkedin (17k+) | Google Scholar | Website |
ML Handbook | Medium (100+ technical blogs featured at top publishers) | Github

PUBLICATIONS

- SERA — an embedding-based tool-routing and parallel-execution framework across **40+** tools — achieved **90%** behavioral similarity to ReAct with **50%** latency drop and **30%** fewer tool failures.
- Architected ROMA, a recursive deep search agent, using a dependency-graph layer for intra-depth subagent communication and parallel tool calling — achieving a **10%** absolute lift over SOTA on the SEAL-0 hierarchical-task benchmark.
- Finetuned AlexNet on defective 3D parts, lifting accuracy to **95%+** via dual hyperparameter tuning — published in AST, Vol. 130.

OPEN SOURCE CONTRIBUTIONS

- Designed TraceDB (cross-iteration memory) for EvoSkill replacing stateless evolution in **continuous self improving harness**, achieving **+7.3%** on OfficeQA and **+5.3%** zero-shot transfer to BrowseComp, proving cross-task generalization.
- Authored a comprehensive ML Handbook; bridging foundations with cutting-edge architectures, utilized by **5,000+** developers.
- Extended **GEPA** with multi-model ensemble generation and LLM-as-judge scoring cutting optimization time by **25%**.

WORK EXPERIENCE

Sentient Labs (Open Source Startup) — **Remote, USA**
AI Research Intern — *Nov 2025 – May 2026*

- Collected branching reward-labeled traces from SERA and used DPO to improve agent task success by **18%** vs the baseline.
- Designed SERA's benchmark and evaluation framework to assess tool-grounded responses on relevance and temporal freshness.
- Implemented automated agentic harness tracing (Harbor ATIF) for Arena; contributions merged into the official Harbor repo.
- Designed MCP Sidecar Architecture for secure agent tool hosting with builder/runtime container split and Caddy reverse proxy.

Strategy — **Virginia, USA**
Software Engineering - AI Research Intern — *June 2025 – Aug 2025*

- Optimized **ReAct**-based Deep Research agent by integrating planners and external tools achieving **40%** quality improvement.
- Built an **Azure Blob connector tool** using RAG architecture, enabling one-click ingestion of **1,000+** files across **10+** formats.
- Built a long-context **LLM evaluator** using LLM-as-a-Judge and checklist scoring, achieving **94%** agreement with human labels.

Piramal Capital & Housing Finance — **India**
Data Scientist - AI Engineer — *Jun 2021 – Aug 2024*

- Led a team of 5 to design a patented RAG-based **Text2SQL Graph AI agent**, leveraging **knowledge graphs**, LLMs, Graph Neural Networks (GNN) and Langchain, securing **$1M+** in management funding for Generative AI initiatives.
- Upgraded unimodal RAG system to a **multimodal RAG** using multimodal VLMs, improving retrieval relevance by **35%**.
- **MLOps/LLMOps**: Deployed open-source LLMs via **CI/CD** pipelines on ECS using **Docker**, reducing build latency by **30%**.

TECHNICAL SKILLS

Languages & Databases	: Python3, SQL, Vector-DB, Pinecone, Neo4j, PostgreSQL, Snowflake, MongoDB, C++,
Frameworks	: PyTorch, Hugging Face, Langchain, LangGraph, LlamaIndex, PySpark, Scikit-Learn, Pandas, NumPy
Dev Tools	: AWS, VS Code, Git/Github, Docker, MLFlow, vLLM, UnSloth, W&B, Rest API, Airflow
AI Specializations	: RAG, LLMs, Post-training, Generative AI, Deep Learning, NLP, CV, Machine Learning, MLOps, Distributed Training, Reinforcement Learning (RL), Inference Optimization (Quantization/Distillation), DSA

EDUCATION

University of Maryland, College Park, MD, USA — **Aug 2024 – May 2026**
Master of Science in Machine Learning; TA for Machine Learning (x2) — *GPA: 3.90 / 4.0*

Birla Institute of Technology and Science (BITS) Pilani — **Aug 2018 – Jun 2022**
Bachelor of Technology in Mechanical Engineering (Minor: ***Data Science****); TA for Linear Optimization* — *GPA: 3.77 / 4.0*

PROJECTS – PAPER IMPLEMENTATIONS

- Implemented DeepSeek-R1 SLM with MHLA, Sparse MoE, KV Caching to achieve **3x** inference speedup and **35%** latency drop.
- Improved multimodal image-text alignment by **27%** using Visual Language Model (VLM) architecture with CLIP encoder.
- Developed **Diffusion Transformers** based video generation model achieving **30%** faster inference than U-Net baselines.

ACHIEVEMENTS & EXTRACURRICULAR

- Selected for **ACM India Summer School** in NLP by IIIT Hyderabad and **Microsoft** with only **0.5%** acceptance rate.
- Accomplished **top 10** rankings in UCMAS (Mental Math) Competition for **two consecutive years**.
- Captained district cricket team while advancing through Art of Living's meditation disciplines and mindfulness practices.